JOY TEACHER User Guide

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Overview

The Desk Watch JOY TEACHER is a small desk console for the teacher. Its two-line, sixteen-character screen shows an exam countdown, the temperature, the humidity and the time. A joystick on top lets you move from one page to another and adjust every function. The readings come from an SHT sensor and the time from an RTC module.

It has eleven menu entries:

- **Exam**: an exam timer that counts the papers handed in (first entry of the menu).
- **Temp & Hum** (shown as "Time Temp Hum" in the menu): temperature and humidity with alerts, over five pages (readings, minimum, maximum, average, automatic scrolling).
- **Timer**: a work timer with breaks (Pomodoro method).
- **Counter, Stopwatch, Alarm, Dice, Chess Clock, Melody**.
- **Settings** and **Sleep mode**.

The RTC module is required. Without it, the Temp & Hum page still works (the time shows "--:--:--"), but the clock, the time setting and the alarm ask you to install the module, and automatic night mode does not work.

The favorite is not a menu entry: it is chosen in Settings and sets the page shown at start-up.



JOY TEACHER: the menu opens on Exam, entry 1 of 11.

Starting up and the joystick

Turning the console on

1. Plug the USB cable into a computer, a charger or a power bank.

2. **Do not touch the joystick** during start-up: the console measures its rest position at that moment.
3. The screen shows DESK WATCH, PR BELAMINE MAKERLAB, then JOY TEACHER with a loading bar. It then checks the modules ("RTC clock OK" or "No RTC clock", "Sensor OK" or "Sensor missing"), then opens the Favorite page. By default, this is the menu.

The console has no power switch: to turn it off, unplug the USB cable.

Joystick gestures

Gesture	Effect
Up or down	Scroll the menu, change a value (held: it repeats, faster after 1.5 s)
Right, or press the head	Confirm (OK)
Right or press, twice within 0.4 s	Double click
Left	Back (menu, cancel)
Left held 2.5 s	Go to sleep

The double click is mostly used to go back to the menu and to reset. Its exact effect is described page by page below.

The menu

From any page, push the joystick left to open the menu. Move up or down to choose an entry, then confirm with right or a press. Left closes the menu and takes you back to the previous page. The screen changes colour depending on the page.

The menu pages

Exam

A timer to keep an eye on the length of a test and to count the papers as they are handed in. It is the first entry of the menu.

Setup, in three steps. Move up or down to change the value, confirm to go to the next step. Left goes back to the previous step; from the first one, it takes you back to the menu. The background is blue during setup.

1. **Exam duration**: from 1 to 240 minutes (35 minutes at the start).
2. **Expected papers**: from 1 to 99 (24 at the start).
3. **EXAM READY**: the screen recalls the duration and the number of papers. Confirm to start.

```
Exam duration
35 min       OK
```

During the exam. The first line shows the time left, the second the papers handed in out of the papers expected. The background is green.

```
EXAM      34:59
Papers     03/24
```

- **Count a paper**: right or press the head, each time a paper is handed in. Counting is immediate, even after time is up; two quick pushes count two papers.
- **Open the actions**: push left. Move up or down, confirm: Back, Subtract 1 paper (in case of a mistake), Pause or Resume, +5 min, End exam, Main menu. A double click closes the actions without changing anything. While paused, the screen shows PAUSED.
- **+5 min** adds five minutes (up to 4 hours in total); if time was up, the countdown starts again.
- **End exam** asks for a confirmation: confirm for yes, double click for no.

The end.

- Time is up: two short beeps, the screen turns red and shows TIME IS UP. You can still count the papers that come in.
- All papers in: the screen turns green and shows ALL PAPERS IN, alternating with TOTAL, the actual working time, pauses excluded.
- After End exam, the screen shows TOTAL. A press takes you back to the setup of a new exam.

> If you open the menu during the exam, the countdown keeps running in the background, and automatic sleep is suspended as long as the session is open. The session is kept in memory (RAM): if you unplug the console, it is lost.

Temp & Hum

The first line shows the time ("--:--:--" if the RTC module is missing), the second line the readings. Push the joystick down or up to change page: two small arrows to the left of the time remind you of this.

- **Page 1, the readings**: live temperature and humidity. The background is turquoise when everything is fine and the OK indicator appears on the right. Alerts appear on this page.
- **Pages 2, 3 and 4**: the bottom line shows the minimum (MIN), the maximum (MAX), then the average (AVG) of temperature and humidity, calculated over the last 30 readings. The arrows stay to the left of the time.
- **Page 5, AUT**: the first four pages scroll by themselves, every 3 seconds. To the left of the time, "AUT" alternates with the arrows.
- When a reading goes outside the thresholds, a message alternates with the readings: TEMP TOO HIGH or LOW, HUM TOO HIGH or LOW. The background turns red for a high threshold and blue for a low one. The buzzer sounds: two short beeps for a high threshold, one long beep for a low one.
- **Right (a single push)**: sets the four thresholds, in the order Tmin, Tmax, Hmin, Hmax (temperature from 10 to 40 °C in steps of 0.5 °C, humidity from 10 to 90 %, with a minimum gap of 1 °C or 5 points between low and high). Move up or down to change the value, confirm to go

to the next threshold. After the last one, the "Choose OK" screen asks you to push right to save. Left goes back to the previous threshold. Default values: 20 and 28 °C, 40 and 60 %.

- If the screen shows "Sensor missing", check the SHT sensor cable.

The favorite is chosen in Settings: Menu, Time Temp Hum, Timer, Stopwatch, Counter, Dice, Chess clock or Exam. By default, it is the menu.

Timer

A work timer with automatic breaks (Pomodoro method).

- When stopped, move up or down to set the work time: from 1 to 60 minutes in steps of 1 minute, then from 65 minutes to 4 hours in steps of 5 minutes.
- Confirm to start. Confirm again to pause or resume.
- After each work period comes a 5-minute break, and a 15-minute break every fourth period. The buzzer announces each change. The background is blue when stopped, orange during work, yellow when paused, green during a short break and cyan during a long break.
- Double click: reset; once at zero, a double click takes you back to the menu.
- If you open the menu during the countdown, the timer keeps running in the background.

Counter

The counter starts in ACTIVE mode. Push up to add 1 (up to 9999) and down to subtract 1 (down to 0): the two arrows on the screen remind you of this. Pressing the joystick head switches to PAUSE and the counter stops moving; press again to reactivate it. Push right to ask for a reset: down = yes, up = no, or do nothing for 5 seconds to cancel (left and double click also cancel).

Stopwatch

Confirm to start, then to stop; confirm again to resume. The time is shown in minutes, seconds and tenths. Double click: reset; once at zero, back to the menu. The stopwatch keeps running if you open the menu.

Alarm

The RTC module is required. When you arrive on the page, the screen shows "UP: Enable alarm" and "DOWN: Disable".

- **Up**: enables the alarm and opens its settings. **Down**: disables the alarm; the screen confirms, then returns to the menu.
- Settings: confirm at each step to move to the next: hours, minutes, melody, volume. The melody and the volume are previewed while you choose. After the volume, the "Alarm OK" screen confirms that it is saved. Left goes back to the previous step.
- Eight ringtones are available: Birthday, Für Elise, Mario Bros, Tetris, Aurora, Cosmic Pulse, Arcade Hero and Epic Rise.
- When the alarm rings, touching the joystick (a direction or a press) stops the ringing for the day and opens the menu; the alarm stays enabled for the next day. With no reaction, the ringing stops after 2 minutes, starts again 5 minutes later, and so on until you touch the joystick.

- The alarm rings even when the console is asleep.

Dice

At the start, the first line shows "DICE ↑↓ sides" and the second "OK=roll": move up or down to choose the number of sides (2, 4, 6, 8, 10, 12, 20, 30 or 100). Each press on the head (or push to the right) rolls the die: "DICE rolling" is displayed, a rattle accompanies the roll for about one second, then the result stays displayed in brackets on the right, with "OK=roll" at the top. Double click: back to the menu.

Chess Clock

Choose the time control when stopped (minutes + increment in seconds): 1+0, 3+0, 3+2, 5+0, 5+3, 10+0, 10+5, 15+10, 20+0, 30+0 or 60+0. Confirm to start; player 1 begins.

- During the game, any movement of the joystick hands over to the other player, with a small beep.
- Press on the head of the joystick (or hold for about 1.2 s): pause. Then confirm to resume, or double click to reset.
- When a player's time runs out, the screen turns red, shows END and the buzzer sounds three times.
- Leaving the page pauses the game.

Melody

Move up or down to choose a tune (Birthday, Für Elise, Mario Bros or Tetris), confirm to play, pause or resume. Double click: back to the menu. The volume is set in Settings.

Settings and sleep mode

Settings

Move up or down to choose a setting, confirm to change it, confirm again to save it. Left or a double click cancels the setting in progress; if there is none, they take you back to the menu.

Setting	Choices
Brightness	8 levels, from 5 % to 100 % (85 % by default), with instant preview
Volume	Five levels: Min, Low, Mid, High or Max (Mid by default)
Boot sound	Yes or no: the start-up melody (off by default)
Auto sleep	Never, 1, 2, 5, 10, 15, 30 minutes or 1 h
Night mode	No, Auto (from 10 pm to 6 am, with the RTC) or Yes: dimmed screen
Set time	Sets the hours then the minutes of the RTC
Favorite	Menu (default), Time Temp Hum, Timer, Stopwatch, Counter, Dice, Chess clock or Exam
Factory	After two confirmations (open, then confirm), restores all the original settings: brightness, volume, sound, sleep, night mode, favorite, thresholds, alarm and timer duration (25 min). The RTC time does not change.

All settings are kept when you unplug the console.

Sleep mode

The last menu entry, or holding left for 2.5 seconds, turns the screen off after "Goodbye". With automatic sleep, the console turns itself off after the chosen delay, except during a chess game in progress or an open exam session.

- To wake the console, move the joystick or press it.
- The alarm and the background functions keep running during sleep.
- Temperature and humidity alerts stay silent during sleep.
- Sleep mode turns the screen off, but the console stays powered: unplug the USB cable to stop it completely.

Troubleshooting

Problem	What to do
The screen stays black	Move the joystick: the console may be asleep. Then check the USB cable, the Grove cable of the screen and the brightness in Settings.
The joystick chooses by itself or seems inverted	It was touched during start-up. Unplug the USB cable, plug it back in, and do not touch the joystick until the Favorite page is displayed.
"Sensor missing"	Check that the SHT sensor is plugged into an I2C socket, then unplug and replug the USB cable.
The time shows "--:--:--" or the alarm asks for the RTC module	The RTC module is not detected: check its cable on an I2C socket.
No sound	Raise the volume (up to Max) and check the buzzer on socket D3. Temperature alerts are silent during sleep and in night mode.
The time is wrong	Use Settings, then Set time.
An alert keeps coming back	Adjust the thresholds: on the Temp & Hum page, push the joystick to the right, so that they suit the room.
A paper was counted by mistake during the exam	Push left to open the actions, choose "Subtract 1 paper" and confirm.
The exam disappeared after a power cut	The session is kept in memory (RAM): it is lost if the console is unplugged. Start a new exam from the Exam page.

If nothing works, note what the screen shows and compare it with the assembly guide.